

Last login: Thu Jun 18 15:16:13 on ttys002
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
psql (18.4)
Type "help" for help.

```
tpch=# SET ROLE alice;
SET
tpch=> EXPLAIN (ANALYZE, VERBOSE, BUFFERS)
SELECT COUNT(*)
FROM (
    SELECT l_orderkey, l_linenum
    FROM lineitem
    WHERE l_orderkey > 2000000
) l_alias;
WARNING:  Log 1: Planner Input:
DETAIL:  {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname l_alias
:colnames <>
}
:eref
{ALIAS
:aliasname l_alias
:colnames ("l_orderkey" "l_linenum")
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum"
"l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnfl
ag" "l_linestatus" "l_shipdate" "l_commitdate" "l_receiptdate"
"l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationk
ey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
```

```

        :fromlist (
            {RANGETBLREF
             :rtindex 1
            }
        )
        :quals
        {FUNCEXP
         :funcid 17005
         :funcresulttype 16
         :funcresultset false
         :funcvariadic false
         :funcformat 0
         :funccollid 0
         :inputcollid 0
         :args (
             {VAR
              :varno 1
              :varattno 6
              :vartype 701
              :vartypmod -1
              :varcollid 0
              :varnullingrels (b)
              :varlevelsup 0
              :varreturningtype 0
              :varnosyn 1
              :varattnosyn 6
              :location -1
             }
         )
         :location -1
        }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
        {TARGETENTRY
         :expr
         {CONST
          :consttype 23
          :consttypmod -1
          :constcollid 0
          :constlen 4
          :constbyval true
          :constisnull false
          :location -1
          :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
         }
         :resno 1
         :resname ?column?
         :ressortgroupref 0
         :resorigtbl 0
         :resorigcol 0
         :resjunk false
        }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :location -1
}
)
)
)
:rteperminfos (
    {RTEPERMISSIONINFO
     :relid 16566
     :inh true
     :requiredPerms 2
     :checkAsUser 0
     :selectedCols (b 0 11)
     :insertedCols (b)
     :updatedCols (b)
    }
)
)
:jointree
{FROMEXPR
 :fromlist (
     {RANGETBLREF
      :rtindex 1
     }
 )
 :quals
 {OEXPR
  :opno 521
  :opfuncid 147
  :opresulttype 16
  :opresultset false
  :opcollid 0
  :inputcollid 0
  :args (
      {VAR
       :varno 1
       :varattno 1
       :vartype 23
       :vartypmod -1
       :varcollid 0
       :varnullingrels (b)
       :varlevelsup 0
       :varreturningtype 0
       :varnosyn 1
       :varattnosyn 1
       :location 123
      }
      {CONST
       :consttype 23
       :consttypmod -1
       :constcollid 0
       :constlen 4
       :constbyval true
       :constisnull false
       :location 136
       :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
      }
  )
  :location 134
 }
 }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
     :expr
     {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
    }
  }
)

```

```

:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 70
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:varatype 23
:varatypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 82
}
:resno 2
:resname l_linenum
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:security_barrier false
:reliid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rtperminfos <>
:jointree
{FROMEXPR
:fromlist (
{RANGEtblREF
rtindex 1
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 28
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 43
}
:resno 1
:resname count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: max_parallel_hazard_walker: FuncExpr funcid=17005 propparallel=s (s=safe r=restricted u=unsafe)

WARNING: Log 2 : After pull up At starting of Path Generation:

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
}

```

```

:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
  :alias
    {ALIAS
      :aliasname s
      :colnames <>
    }
  :eref
    {ALIAS
      :aliasname s
      :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
        s_acctbal" "s_comment")
    }
  :rtekind 0
  :relid 16541
  :inh false
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
  :relid 16541
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 13)
  :insertedCols (b)
  :updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
  :rtindex 1
}
)
:quals (
  {FUNCEXP
  :funcid 17005
  :funcresulttype 16
  :funcresultset false
  :funcvariadic false
  :funcformat 0
  :funccollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 6
    :vartype 701
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
    }
  )
  :location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {FUNCEXP :funcid 17005 :funcresulttype 16 :funcresultset false :funcvariadic false :funcformat 0 :funccollid 0 :inputcollid 0 :args ({VAR :v
arno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1}) :location -1}) :is_pushed_down true :can_join f
alse :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :oute
r_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :out
er_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {FUNCEXP :funcid 17005 :funcresulttype 16 :funcresultset false :funcvariadic false :funcformat 0 :funccollid 0 :inputcollid 0 :args ({VAR :
varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1}) :location -1}) :is_pushed_down true :can_join
false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :out
er_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :out
er_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPA
RALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 3: At the starting of Plan creation:
DETAIL:
{PATH
:pathtype 339
:parent_relds (b 1)
:pathtarget
{PATHTARGET
:expr <>
:sortgrouprefs ()
:cost.startup 0
:cost.per_tuple 0
:width 0
:has_volatile_expr 0
}
}
:required_outer (b)
:parallel_aware false
:parallel_safe true
:parallel_workers 0
:rows 3333

```

```

:disabled_nodes 0
:startup_cost 0
:total_cost 2818
:pathkeys <>
}

WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname l_alias
:colnames <>
}
:eref
{ALIAS
:aliasname l_alias
:colnames ("l_orderkey" "l_linenumber")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesstatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0
:location -1
}
))
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 2
}
)
:quals (
{OPEXPR
:opno 821
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 123
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 136
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location 134
}
)
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{AGGREG
:aggfnoid 2803

```

```

:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 43
}
:resno 1
:resname count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:smt_location 0
:smt_len 0
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRUCTINFO :clause {PARAM :paramkind 1 :paramid 0 :paramtype 16 :paramtypmod -1 :paramcollid 0 :location -1} :is_pushed_down true :can_join false :pseudocons
tant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 0 :clause_relids (b 2) :incompatible_relids (b) :outer_relids (b) :left
_relids (b) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :
hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRUCTINFO :clause {OEXPR :opno 521 :opfuncid 147 :opre
suldtype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattn
osyn 1 :location -1} {CONST :constype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [-128 -124 30 0 0 0 0 ]}) :location -1} :is_pushed_
down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relids (b 2) :required_relids (b 2) :incompati
ble_relids (b) :outer_relids (b) :left_relids (b 2) :right_relids (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_e
m <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=0 in_safe_list=1 → reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRUCTINFO :clause {PARAM :paramkind 1 :paramid 0 :paramtype 16 :paramtypmod -1 :paramcollid 0 :location -1} :is_pushed_down true :can_join false :pseudocon
stant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 0 :clause_relids (b) :required_relids (b 2) :incompatible_relids (b) :outer_relids (b) :lef
t_relids (b) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :
hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRUCTINFO :clause {OEXPR :opno 521 :opfuncid 147 :opr
esulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varatt
nosyn 1 :location -1} {CONST :constype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [-128 -124 30 0 0 0 0 ]}) :location -1} :is_pushed_
_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relids (b 2) :required_relids (b 2) :incompat
ible_relids (b) :outer_relids (b) :left_relids (b 2) :right_relids (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_
em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe =
true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({AGGREF :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilt
er <> :aggstar true :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({AGGREF :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfil
em <> :aggstar true :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 3: At the starting of Plan creation:
DETAIL: {AGGPATH
:path.pathtype 365
:parent_relids (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 1
:path.disabled_nodes 0
:path.startup_cost 93401.31968373689
:path.total_cost 93401.32968373688
:path.pathkeys <>
:subpath
{GATHERPATH
:path.pathtype 368
:parent_relids (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 2
:path.disabled_nodes 0
:path.startup_cost 92400.25919918844
:path.total_cost 92400.26919918843
:path.pathkeys <>
:subpath
{AGGPATH
:path.pathtype 365
:parent_relids (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe true
:path.parallel_workers 2
:path.rows 1
:path.disabled_nodes 0
:path.startup_cost 92400.25919918844
:path.total_cost 92400.26919918843
:path.pathkeys <>
:subpath
{INDEXPATH
:path.pathtype 342
:parent_relids (b 2)
:path.pathtarget
{PATHTARGET
:exprs <>
:sortgrouprefs <>
:cost.startup 0
:cost.per_tuple 0
:width 0
:has_volatile_expr 0
}
:required_outer (b)
:path.parallel_aware true
:path.parallel_safe true
:path.parallel_workers 2
:path.rows 838673
:path.disabled_nodes 0
:path.startup_cost 0.4325
:path.total_cost 90323.57669918844
:path.pathkeys <>
:indexinfo
{INDEXOPTINFO
```

```

:indexoid 17007
:reltablespace 0
:pages 16457
:tuples 6001215
:tree_height 2
:ncolumns 2
:nkeycolumns 2
:indexkeys ( 1 4)
:indexcollations ( 0 0)
:opfamily ( 1976 1976)
:opcintype ( 23 23)
:sortopfamily ( 1976 1976)
:reverse_sort ( false false)
:nulls_first ( false false)
:canreturn ( true true)
:relam 403
:indpred <>
:indextlist (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:indrestrictinfo (
{RESTRICTINFO
:clause
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0
:location -1
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof false
:has_volatile 0
:security_level 0
:num_base_rels 0
:clause_relids (b)
:required_relids (b 2)
:incompatible_relids (b)
:outer_relids (b)
:left_relids (b)
:right_relids (b)
:orclause <>
:rinfo_serial 1
:eval_cost.startup 0
:eval_cost.per_tuple 0
:norm_selec 0.5
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}
{RESTRICTINFO
:clause
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 123
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 136
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location 134
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false

```

```

:leakproof true
:has_volatile 0
:security_level 1
:num_base_rels 1
:clause_relids (b 2)
:required_relids (b 2)
:incompatible_relids (b)
:outer_relids (b)
:left_relids (b 2)
:right_relids (b)
:orclause <>
:rinfo_serial 2
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 0.6644041354651555
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}
):predOK false
:unique false
:nullsnotdistinct false
:immediate true
:hypothetical false
:amcanorderbyop false
:amoptionalkey true
:amsearcharray true
:amsearchnulls true
:amhasgettupl true
:amhasgetbitmap true
:amcanparallel true
:amcanmarkpos true
}
:indexclauses (
{INDEXCLAUSE
:rinfo
{RESTRICTINFO
:clause
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 123
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 136
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location 134
}
):is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof true
:has_volatile 0
:security_level 1
:num_base_rels 1
:clause_relids (b 2)
:required_relids (b 2)
:incompatible_relids (b)
:outer_relids (b)
:left_relids (b 2)
:right_relids (b)
:orclause <>
:rinfo_serial 2
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 0.6644041354651555
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}
):indexquals (
{RESTRICTINFO
:clause
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 123
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 136
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
}
)

```

```

    }
    )
    :location 134
  }
  :is_pushed_down true
  :can_join false
  :pseudoconstant false
  :has_clone false
  :is_clone false
  :leakproof true
  :has_volatile 0
  :security_level 1
  :num_base_rels 1
  :clause_relds (b 2)
  :required_relds (b 2)
  :incompatible_relds (b)
  :outer_relds (b)
  :left_relds (b 2)
  :right_relds (b)
  :orclause <>
  :rinfo_serial 2
  :eval_cost.startup 0
  :eval_cost.per_tuple 0.0025
  :norm_selec 0.6644041354651555
  :outer_selec -1
  :mergeopfamilies <>
  :left_em <>
  :right_em <>
  :outer_is_left false
  :hashjoinoperator 0
  :left_bucketsize -1
  :right_bucketsize -1
  :left_mcvfreq -1
  :right_mcvfreq -1
  :left_hasheqoperator 0
  :right_hasheqoperator 0
  }
  )
  :lossy false
  :indexcol 0
  :indexcols <>
  }
  )
  :indexorderbys <>
  :indexorderbycols <>
  :indexscandir 1
  :indextotalcost 73644.6725
  :indexselectivity 0.6644041354651555
  }
  :aggstrategy 0
  :aggsplit 6
  :numGroups 1
  :transitionSpace 0
  :groupClause <>
  :qual <>
  }
  :single_copy false
  :num_workers 2
  }
  :aggstrategy 0
  :aggsplit 9
  :numGroups 1
  :transitionSpace 0
  :groupClause <>
  :qual <>
  }
  }

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL: {PLANNEDTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded true
:jitFlags 0
:planTree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 93401.31968373689
:plan.total_cost 93401.32968373688
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 20
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 9
:aggno 0
:aggtransno 0
:location 43
}
:resno 1
:resname count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:plan.qual <>
:plan.lefttree

```

```

{GATHER
:plan.disabled_nodes 0
:plan.startup_cost 93400.25919918844
:plan.total_cost 93400.46919918843
:plan.plan_rows 2
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 1
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 20
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 92400.25919918844
:plan.total_cost 92400.26919918843
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 6
:aggno 0
:aggtransno 0
:location 43
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{INDEXONLYSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0.4325
:scan.plan.total_cost 90323.57669918844
:scan.plan.plan_rows 830673
:scan.plan.plan_width 0
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 3
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno -3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:scan.plan.qual (
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paratypmod -1
:paramcollid 0
:location -1
}
)
:scan.plan.lefttree <>

```

```

:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0 1)
:scan.plan.allParam (b 0 1)
:scan.scanreliid 2
:indexid 17007
:indexqual (
  {OPEXPR
    :opno 521
    :opfuncid 147
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno -3
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 1
        :location 123
      }
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location 136
        :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
      }
    )
    :location 134
  }
)
:recheckqual (
  {OPEXPR
    :opno 521
    :opfuncid 147
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno -3
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 1
        :location 123
      }
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location 136
        :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
      }
    )
    :location 134
  }
)
:indexorderby <>
:indextlist (
  {TARGETENTRY
    :expr
    {VAR
      :varno 2
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 2
      :varattno 4
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 4
      :location -1
    }
    :resno 2
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:indexorderdir 1
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0 1)
:plan.allParam (b 0 1)
:aggstrategy 0
:aggssplit 6
:numCols 0
:grpCollIdx ()
:grpOperators ()
:grpCollations ()
:numGroups 1
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:num_workers 2

```

```

:rescan_param 1
:single_copy false
:invisible false
:initParam (b 0)
}
:plan.righttree <>
:plan.initPlan (
{SUBPLAN
:sublinkType 0
:testexpr <>
:paramIds <>
:plan_id 1
:plan_name InitPlan\ 1
:firstColType 2278
:firstColTypeMod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe true
:setParam (i 0)
:parParam <>
:args <>
:startup_cost 0
:per_call_cost 0.8454845484548454
}
)
:plan.extParam (b)
:plan.allParam (b 0)
:aggstrategy 0
:aggssplit 9
:numCols 0
:grpColIdx ()
:grpOperators ()
:grpCollations ()
:numGroups 1
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname l_alias
:colnames <>
}
:eref
{ALIAS
:aliasname l_alias
:colnames ("l_orderkey" "l_linenumbr")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumbr" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:unprunableRelids (b 2 3)
:permInfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
)
:resultRelations <>
:appendRelations <>
:subplans (
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 2818
:scan.plan.plan_rows 3333
:scan.plan.plan_width 0
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 4
:scan.plan.targetlist <>
:scan.plan.qual (
{FUNCEXP
:funcid 17005

```

```

:funcresulttype 16
:funcresultset false
:funcvariadic false
:funcformat 0
:funccollid 0
:inputcollid 0
:args (
  {VAR
    :varno 3
    :varattno 6
    :vartype 701
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 6
    :location -1
  }
)
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 3
}
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids (o 16566 16541)
:invalidItems (
  {PLANINWALITEM
    :cacheId 47
    :hashValue 2210181313
  }
)
:paramExecTypes (o 16 0)
:utilityStat <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: Log 1: Planner Input:

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStat <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree
  {FROMEXPR
    :fromlist <>
    :quals <>
  }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {OEXPR
      :opno 674
      :opfuncid 297
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {PARAM
          :paramkind 0
          :paramid 1
          :paramtype 701
          :paramtypmod -1
          :paramcollid 0
          :location 0
        }
      )
      {FUNCEXP
        :funcid 316
        :funcresulttype 701
        :funcresultset false
        :funcvariadic false
        :funcformat 2
        :funccollid 0
        :inputcollid 0
        :args (
          {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location 10
            :constvalue 4 [ -24 3 0 0 0 0 0 0 ]
          }
        )
      }
    }
    :location -1
  }
)
:location 8
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
}

```

```

:stmt_len 0
}

WARNING: max_parallel_hazard_walker: Param paramid=1 in_safe_list=0 → reason=outer_reference
WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 propparallel=s (s=safe r=restricted u=unsafe)
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {
  QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs false
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSublinks false
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity false
  :hasGroupRTE false
  :isReturn false
  :cteList <>
  :rtable (
    {RANGETBENTRY
      :alias <>
      :eref
        {ALIAS
          :aliasname *RESULT*
          :colnames <>
        }
      :rtekind 8
      :lateral false
      :inFromCl false
      :securityQuals <>
    }
  )
  :rteperminfos <>
  :jointree
    {FROMEXPR
      :fromlist (
        {RANGETBLREF
          :rtindex 1
        }
      )
      :quals <>
    }
  :mergeActionList <>
  :mergeTargetRelation 0
  :mergeJoinCondition <>
  :targetList (
    {TARGETENTRY
      :expr
        {OEXPR
          :opno 674
          :opfuncid 297
          :opresulttype 16
          :opretset false
          :opcollid 0
          :inputcollid 0
          :args (
            {PARAM
              :paramkind 0
              :paramid 1
              :paramtype 701
              :paramtypmod -1
              :paramcollid 0
              :location 0
            }
            {CONST
              :consttype 701
              :consttypmod -1
              :constcollid 0
              :constlen 8
              :constbyval true
              :constisnull false
              :location -1
              :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
            }
          )
          :location 8
        }
      :resno 1
      :resname ?column?
      :ressortgroupref 0
      :resorigtbl 0
      :resorigcol 0
      :resjunk false
    }
  )
  :override 0
  :onConflict <>
  :returningOldAlias <>
  :returningNewAlias <>
  :returningList <>
  :groupClause <>
  :groupDistinct false
  :groupingSets <>
  :havingQual <>
  :windowClause <>
  :distinctClause <>
  :sortClause <>
  :limitOffset <>
  :limitCount <>
  :limitOption 0
  :rowMarks <>
  :setOperations <>
  :constraintDecls <>
  :withCheckOptions <>
  :stmt_location 0
  :stmt_len 0
}

WARNING: is_parallel_safe: INPUT qual = ({OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({PARAM :paramkind 0 :paramid 1 :paramtype 701 :paramtypmod -1 :paramcollid 0 :location -1}) {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]}) :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({PARAM :paramkind 0 :paramid 1 :paramtype 701 :paramtypmod -1 :paramcollid 0 :location -1}) {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]}) :location -1}) |
parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 3: At the starting of Plan creation:
DETAIL: {
  PROJECTIONPATH
  :path.pathtype 331
  :parent_relids (b 1)
  :required_outer (b)
  :path.parallel_aware false
  :path.parallel_safe false
  :path.parallel_workers 0
  :path.rows 1
  :path.disabled_nodes 0
  :path.startup_cost 0
  :path.total_cost 0.0125
  :path.pathkeys <>
  :subpath
    {GROUPRESULTPATH
      :path.pathtype 331
      :parent_relids (b 1)
      :path.pathtarget
        {PATHTARGET
          :exprs <>
          :sortgrouprefs <>
          :cost.startup 0
          :cost.per_tuple 0
          :width 0
          :has_volatile_expr 0
        }
      :required_outer (b)
      :path.parallel_aware false
      :path.parallel_safe false
    }
  }
}

```

```

:plan.parallel_workers 0
:plan.rows 1
:plan.disabled_nodes 0
:plan.startup_cost 0
:plan.total_cost 0.01
:plan.pathkeys <>
:quals <>
}
:dummypp true
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL: {PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded false
:jitFlags 0
:planTree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 92400.25919918844
:plan.total_cost 92400.26919918843
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranspype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 6
:aggno 0
:aggtransno 0
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{INDEXONLYSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0.4325
:scan.plan.total_cost 90323.57669918844
:scan.plan.plan_rows 830673
:scan.plan.plan_width 0
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 3
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -3
:varattno 1
:varstype 23
:varstypepm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno -3
:varattno 2
:varstype 23
:varstypepm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypepm -1
:paramcollid 0
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0 1)
:scan.plan.allParam (b 0 1)
:scan.scanrelid 2
:indexid 17007
:indexqual (
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false

```

```

:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno -3
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
  }
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
  }
)
:location -1
}
)
:recheckqual (
  {OPEXPR
    :opno 521
    :opfuncid 147
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno -3
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 1
        :location -1
      }
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
      }
    )
    :location -1
  }
)
:indexorderby <>
:indextlist (
  {TARGETENTRY
    :expr
    {VAR
      :varno 2
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 2
      :varattno 4
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 4
      :location -1
    }
    :resno 2
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:indexorderdir 1
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0 1)
:plan.allParam (b 0 1)
:aggstrategy 0
:aggssplit 0
:numCols 0
:grpColldx <>
:grpOperators <>
:grpCollations <>
:numGroups 1
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:rtable (
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname l_alias
      :colnames <>
    }
  }
  :eref
  {ALIAS
    :aliasname l_alias
    :colnames ("l_orderkey" "l_linenumber")
  }
)
:rtekind 1
:subquery <>
:security_barrier false
:reliid 0

```

```

:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_qualified" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:unprunableRelids (b 2 3)
:permInfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
)
:resultRelations <>
:appendRelations <>
:subplans (
{SESCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 2818
:scan.plan.plan_rows 3333
:scan.plan.plan_width 0
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 4
:scan.plan.targetlist <>
:scan.plan.qual (
{FUNDEXPR
:funcid 17005
:funcresulttype 16
:funcresultset false
:funcvariadic false
:funcformat 0
:funccollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 6
:varstype 701
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 6
:location -1
}
)
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 3
}
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids <>
:invalidItems <>
:paramExecTypes (o 16 0)
:utilityStmt <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL: {PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded false
:jitFlags 0
:planTree
{AGG
:plan.disabled_nodes 0

```

```

:plan.startup_cost 92400.25919918844
:plan.total_cost 92400.26919918843
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
  {TARGETENTRY
    :expr
      {AGGREF
        :aggfnoid 2803
        :aggtype 20
        :aggcollid 0
        :inputcollid 0
        :aggtranstype 20
        :aggargtypes <>
        :aggdirectargs <>
        :args <>
        :aggorder <>
        :aggdistinct <>
        :aggfilter <>
        :aggstar true
        :aggvariadic false
        :aggkind n
        :aggpresorted false
        :agglevelsup 0
        :aggsplit 6
        :aggno 0
        :aggtransno 0
        :location -1
      }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:plan.qual <>
:plan.lefttree
  {INDEXONLYSCAN
    :scan.plan.disabled_nodes 0
    :scan.plan.startup_cost 0.4325
    :scan.plan.total_cost 90323.57669918844
    :scan.plan.plan_rows 830673
    :scan.plan.plan_width 0
    :scan.plan.parallel_aware true
    :scan.plan.parallel_safe true
    :scan.plan.async_capable false
    :scan.plan.plan_node_id 3
    :scan.plan.targetlist (
      {TARGETENTRY
        :expr
          {VAR
            :varno -3
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 2
            :varattnosyn 1
            :location -1
          }
        :resno 1
        :resname <>
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
      }
    )
    :scan.plan.qual (
      {PARAM
        :paramkind 1
        :paramid 0
        :paramtype 16
        :paramtypmod -1
        :paramcollid 0
        :location -1
      }
    )
    :scan.plan.lefttree <>
    :scan.plan.righttree <>
    :scan.plan.initPlan <>
    :scan.plan.extParam (b 0 1)
    :scan.plan.allParam (b 0 1)
    :scan.scanrelid 2
    :indexid 17007
    :indexqual (
      {OPEXPR
        :opno 521
        :opfuncid 147
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno -3
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 2
            :varattnosyn 1
            :location -1
          }
        )
        {CONST
          :consttype 23
          :consttypmod -1
          :constcollid 0
          :constlen 4
          :constbyval true
          :constisnull false
          :location -1
          :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
        }
      }
    )
  }
)

```

```

    )
    :location -1
  }
)
:recheckqual (
  {OPEXPR
  :opno 521
  :opfuncid 147
  :opresulttype 16
  :opreset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno -3
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
    }
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
    }
  )
  :location -1
)
)
:indexorderby <>
:indextlist (
  {TARGETENTRY
  :expr
  {VAR
  :varno 2
  :varattno 1
  :vartype 23
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 2
  :varattnosyn 1
  :location -1
  }
  :resno 1
  :resname <>
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 2
  :varattno 4
  :vartype 23
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 2
  :varattnosyn 4
  :location -1
  }
  :resno 2
  :resname <>
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
  }
  }
)
:indexorderdir 1
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0 1)
:plan.allParam (b 0 1)
:aggstrategy 0
:aggsplit 6
:numCols 0
:grpColidx <>
:grpOperators <>
:grpCollations <>
:numGroups 1
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:rtable (
  {RANGETBENTRY
  :alias
  {ALIAS
  :aliasname l_alias
  :colnames <>
  }
  :eref
  {ALIAS
  :aliasname l_alias
  :colnames ("l_orderkey" "l_linenum")
  }
  :rtekind 1
  :subquery <>
  :security_barrier false
  :relid 0
  :inh false
  :relkind <>
  :rellockmode 0
  :perminfoindex 0
  :lateral false
  :inFromCl true
  :securityQuals <>
  }
  {RANGETBENTRY
  :alias <>
  :eref
  {ALIAS
  :aliasname lineitem
  :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
  }
  :rtekind 0
  :relid 16566
  :inh false
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false

```

```

:inFromCl true
:securityQuals <>
}
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:unprunableRelids (b 2 3)
:permInfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
)
:resultRelations <>
:appendRelations <>
:subplans (
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 2818
:scan.plan.plan_rows 3333
:scan.plan.plan_width 0
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 4
:scan.plan.targetlist <>
:scan.plan.qual (
(FUNCEXPR
:funcid 17005
:funcresulttype 16
:funcresultset false
:funcvariadic false
:funcformat 0
:funccollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 6
:vartype 701
:vartymod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 6
:location -1
}
)
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 3
}
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids <>
:invalidItems <>
:paramExecTypes (o 16 0)
:utilityStmt <>
:stmt_location -1
:stmt_len -1
}

```

QUERY PLAN

```

Finalize Aggregate  (cost=93401.32..93401.33 rows=1 width=8) (actual time=225.466..226.904 rows=1.00 loops=1)
Output: count(*)
Buffers: shared hit=11200
InitPlan 1
-> Seq Scan on public.supplier s  (cost=0.00..2818.00 rows=3333 width=0) (actual time=3.097..3.097 rows=1.00 loops=1)
Filter: check_supplier_balance_safe(s.s_acctbal)
Buffers: shared hit=211
-> Gather  (cost=93400.26..93400.47 rows=2 width=8) (actual time=225.359..226.896 rows=3.00 loops=1)
Output: (PARTIAL count(*))
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=11200
-> Partial Aggregate  (cost=92400.26..92400.27 rows=1 width=8) (actual time=216.907..216.907 rows=1.00 loops=3)
Output: PARTIAL count(*)
Buffers: shared hit=10989
Worker 0:  actual time=213.519..213.519 rows=1.00 loops=1
Buffers: shared hit=3618
Worker 1:  actual time=215.105..215.105 rows=1.00 loops=1
Buffers: shared hit=3647
-> Parallel Index Only Scan using l_pk on public.lineitem  (cost=0.43..90323.58 rows=830673 width=0) (actual time=0.157..165.543 rows=1333572.67 loops=3)
Output: lineitem.l_orderkey, lineitem.l_linenumber
Index Cond: (lineitem.l_orderkey > 2000000)
Filter: (initplan 1).col1
Heap Fetches: 0
Index Searches: 1
Buffers: shared hit=10989
Worker 0:  actual time=0.052..162.853 rows=1317079.00 loops=1
Buffers: shared hit=3618
Worker 1:  actual time=0.072..164.032 rows=1327848.00 loops=1
Buffers: shared hit=3647

Planning:
Buffers: shared hit=101
Planning Time: 8.328 ms
Execution Time: 239.807 ms
(34 rows)

tpch=>

```